Clustering Model Visualization

Unlike supervised learning models, **clustering models do not have ground-truth labels** to directly calculate accuracy. Therefore, visualization techniques are very important for understanding whether the clusters formed by the model are meaningful and well separated.

The three main visualization techniques used to evaluate clustering models are:

1. **Elbow Method**
2. **Silhouette Coefficient / Silhouette Plot**
3. **Inter-Cluster Distance Maps**

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1. Elbow Method

The **Elbow Method** is mainly used to determine the **optimal number of clusters (k)** in clustering algorithms such as **K-Means**.

Within-Cluster Sum of Squares (WCSS)

The Elbow Method uses **WCSS**, also called **inertia**.

WCSS measures how closely the data points within a cluster are grouped around their cluster centroid.

Basic Idea

The clustering algorithm is run multiple times using different values of k.

For example:

$$k = 1,2,3,4,5, \dots$$

For each value of k, the corresponding WCSS value is calculated and plotted.

- **X-axis:** Number of clusters (k)
- **Y-axis:** WCSS or Inertia

Interpretation of the Elbow Plot

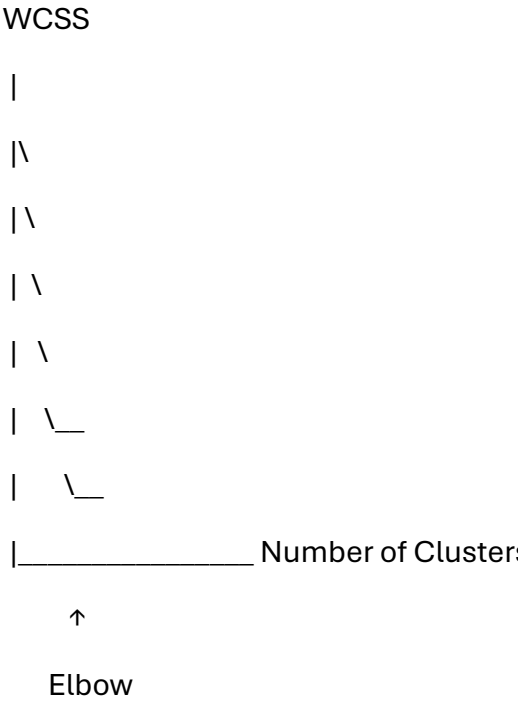
As the number of clusters increases:

$$k \uparrow \ \Rightarrow \ WCSS \downarrow$$

This happens because dividing data into more clusters usually makes each cluster smaller and more compact.

However, after a certain value of k, the decrease in WCSS becomes much slower.

This creates an **elbow-like bend** in the graph.



The value of $k$at the elbow is generally considered the **optimal number of clusters**.

Example

Suppose the WCSS values are:

Number of Clusters (k)	**WCSS**
1	1000
2	600
3	300
4	250
5	220
6	200

The WCSS decreases rapidly from $k = 1$to $k = 3$, but after $k = 3$, the improvement becomes smaller.

Therefore:

$$\boxed{k = 3}$$

may be selected as the optimal number of clusters.

Limitation

The elbow is not always clearly visible.

Sometimes the graph decreases smoothly without showing a clear bend. In such cases, other techniques such as the **Silhouette Coefficient** are useful.

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2. Silhouette Coefficient

The **Silhouette Coefficient** measures how well each data point fits into its assigned cluster.

It considers two important distances:

- a= Average distance between a point and other points in its own cluster.
- b= Average distance between that point and points in the nearest neighbouring cluster.

The silhouette value is calculated as:

$$s = \frac{b - a}{\max(a, b)}$$

The value ranges from:

$$-1 \text{ to } +1$$

Interpretation of Silhouette Values

$$\boldsymbol{s \approx +1}$$

A value close to **+1** indicates that:

- The point is close to points in its own cluster.
- The point is far from neighbouring clusters.
- The clustering assignment is very good.

$$\boldsymbol{s \approx 0}$$

A value close to **0** indicates that:

- The point lies near the boundary between two clusters.
- The clusters may overlap.

$$s < 0$$

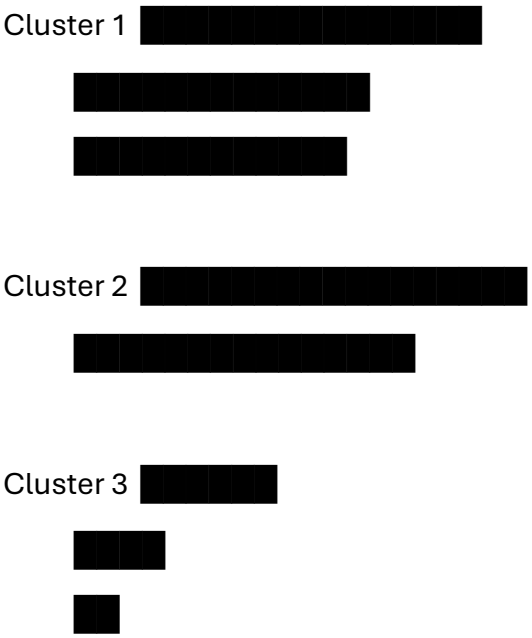
A negative value indicates that:

- The point may be closer to another cluster than its assigned cluster.
- The point may be incorrectly clustered.

Silhouette Plot

A Silhouette Plot displays the silhouette values of data points, usually grouped according to their clusters.

Silhouette Score



Good Clustering

A good silhouette plot usually has:

- Wide bars
- High silhouette values
- Similar quality across clusters
- Few or no negative values

This indicates that the clusters are:

- Compact internally
- Well separated from each other

Poor Clustering

Poor clustering may show:

- Thin bars
- Many values near zero
- Negative silhouette values
- Large differences between clusters

This indicates:

- Overlapping clusters
- Poor cluster separation
- Possible incorrect assignment of points

Why Silhouette Analysis is Useful

The Elbow Method provides a **single overall measure** based on WCSS.

The Silhouette Plot provides a more detailed analysis because it shows:

- Quality of each cluster
- Quality of individual data points
- Possible overlapping clusters
- Possible incorrectly assigned points

Therefore, it provides a more precise evaluation of clustering quality.

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3. Inter-Cluster Distance Maps

An **Inter-Cluster Distance Map** provides a visual representation of how far apart different clusters are from one another.

Since clustering data may have many dimensions, the cluster centers are often projected into **two-dimensional space** using techniques such as:

- **Multidimensional Scaling (MDS)**

Representation

Clusters are usually displayed as circles.

Circle Size

The size of each circle represents:

Number of data points in the cluster

A larger circle means that the cluster contains more observations.

Distance Between Circles

The distance between circles represents how different the clusters are.

- Circles far apart → clusters are well separated.
- Circles close together → clusters are similar or may overlap.

Cluster 1

( )

Cluster 2

( )

Cluster 3

( )

Interpretation

Well-Separated Clusters

If circles are far apart, it indicates:

- Strong separation

- Distinct groups
- Good clustering structure

Overlapping or Nearby Clusters

If circles overlap or are very close:

- Clusters may be similar.
- Cluster boundaries may be unclear.
- The selected number of clusters may need reconsideration.

Advantages

Inter-cluster distance maps provide a simple and intuitive global view of clustering results.

They are particularly useful when explaining clustering results to **non-technical users**, because people can easily understand:

- Which clusters are large
- Which clusters are similar
- Which clusters are well separated

without analysing raw mathematical distance values.

Comparison of Clustering Visualization Techniques

Visualization Technique	Main Purpose	What to Look For
Elbow Method	Select the optimal number of clusters	Clear elbow or bend in the WCSS curve
Silhouette Plot	Evaluate cluster quality and separation	High and consistent silhouette values
Inter-Cluster Distance Map	Visualize overall cluster relationships	Well-separated, non-overlapping clusters

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Key Difference Between the Three

Elbow Method asks:

How many clusters should we create?

Silhouette Analysis asks:

How well does each data point and cluster fit the clustering structure?

Inter-Cluster Distance Map asks:

How are the clusters positioned relative to each other?

Final Exam Answer

Clustering models are evaluated differently from supervised learning models because they do not have ground-truth labels for calculating accuracy. Therefore, visualization techniques are essential for analysing cluster quality and structure. The **Elbow Method** plots WCSS against different values of $k$and selects the number of clusters at the point where the curve forms an elbow. The **Silhouette Coefficient** measures how similar a data point is to its own cluster compared with neighbouring clusters, with values closer to $+1$indicating good clustering and negative values indicating poor assignment. **Inter-Cluster Distance Maps** visually display the distance between clusters, where well-separated circles indicate distinct clusters and overlapping circles may indicate similar or poorly separated clusters. Together, these techniques provide a comprehensive evaluation of clustering quality.